

과제1

- 작은 과제
 - 분량 자유 → attack, 논리적 사유 방식 확인이 목적
 - 출처 표기 자유
 - 4/7 00:00 마감 → 4/6까지 완성해야함
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Further study 01

- **Assignment Summary**
 - 1. Objective:** Resolve a question of personal interest through self-directed additional study.
 - 2. Topic:** Choose **any topic related to material covered in class** and investigate it in depth.
 - 3. Scope:** Select a (related) topic from the content covered **up to Slide 2: Training**.
- **Suggested Approach**
 - Use **internet search**, **AI tools** such as ChatGPT, and **toolkits** such as PyTorch
 - Explore the **theoretical background**
 - Perform **programming-based verification**
- **Required Submission**

Report in pdf format: filename convention= A_01_(yourname).pdf
- The report may be organized freely, but it should include:
 - **Research background**
 - **Theoretical development**
 - **Programming-based validation (you can use AI (e.g., ChatGPT) for programming)**
 - **Conclusion**

- **Appendix**
- **Chatting log with AI (To see how your mind flows)**
- **Program code (can be manual coding or AI-generated code)**
- **Additional results and visualization**
- **Good Topic Candidates**

Since the scope is limited to **Training**, strong topics would be:

 - Why cross-entropy is preferred over MSE for classification
 - How backpropagation is implemented in PyTorch autograd
 - Vanishing gradient and exploding gradient during training
 - The role of initialization in neural network training
 - Why mini-batch SGD works better than full-batch gradient descent in practice
 - Difference in optimizers including modern variations
- **Recommended Report Structure**

Title *Why Cross-Entropy Loss is Preferred over Mean Squared Error for Classification*

1. Research Background

 - Classification is a core supervised learning problem
 - Loss design strongly affects optimization behavior and prediction quality
 - Although MSE can be used, cross-entropy is the standard choice

2. Theoretical Development

 - Definition of classification and probabilistic output
 - Softmax output interpretation
 - MSE formulation
 - Cross-entropy formulation
 - Gradient behavior comparison
 - Why cross-entropy provides a more suitable training signal

3. Programming Verification

- Train the same classifier on the same dataset
 - Compare:
 - training loss curve
 - validation accuracy
 - convergence speed
 - gradient behavior
 - Use PyTorch for implementation
 - **4. Conclusion**
 - Summarize theoretical and empirical differences
 - Explain why cross-entropy is the practical standard for classification
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